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Risks and Bayes Risks (Problems 1–2)

Problem 1

Work through the example we did in class. The two-sample problem with estimation of the difference between Bernoulli probabilities and independent Beta priors with squared error loss. Find the risk function and the Bayes risk for each of the two decision rules.

SOLUTION:

The class example, assembled (notes, Examples 1.8.3–1.8.4): within each arm the outcomes are independent Bernoulli variables with a common expectation, so the arm totals are $X_T \sim \text{Binomial}(n_T, \theta_T)$ and $X_P \sim \text{Binomial}(n_P, \theta_P)$, independent; the parameter is $\theta = (\theta_T, \theta_P) \in [0, 1]^2$; the target is the difference $g(\theta) = \theta_T - \theta_P$ under squared-error loss; the priors are independent $\text{Beta}(1, 1)$ —uniform—on each arm’s parameter. The two decision rules are the difference of the observed rates and the difference of the shaded rates,

$$\hat{\delta} = \frac{X_T}{n_T} - \frac{X_P}{n_P}, \quad \tilde{\delta} = \frac{1 + X_T}{2 + n_T} - \frac{1 + X_P}{2 + n_P}.$$

Everything below is the bias–variance decomposition (notes, Supplement 1.A.1) applied twice, once per rule, followed by an integration against the priors.

For the rate difference, each arm's rate is unbiased and the arms are independent:

$$E^\theta\{\hat{\delta}\} = \frac{n_T\theta_T}{n_T} - \frac{n_P\theta_P}{n_P} = \theta_T - \theta_P \quad (\text{linearity; binomial mean})$$

$$b_{\hat{\delta}}(\theta) = 0 \quad (\text{unbiased})$$

$$\text{Var}^\theta(\hat{\delta}) = \frac{\theta_T(1-\theta_T)}{n_T} + \frac{\theta_P(1-\theta_P)}{n_P} \quad (\text{independent arms; scaling; binomial variance})$$

$$R_{\hat{\delta}}(\theta) = \boxed{\frac{\theta_T(1-\theta_T)}{n_T} + \frac{\theta_P(1-\theta_P)}{n_P}} \quad (\text{bias-variance decomposition})$$

For the shaded difference, work one arm at a time. Writing $T = (1 + X)/(2 + n)$ for an arm with $X \sim \text{Binomial}(n, \theta)$,

$$E^\theta\{T\} = \frac{1 + n\theta}{n + 2}, \quad b_T(\theta) = \frac{1 + n\theta}{n + 2} - \theta = \frac{1 - 2\theta}{n + 2} \quad (\text{linearity; algebra})$$

$$\text{Var}^\theta(T) = \frac{n\theta(1-\theta)}{(n+2)^2} \quad (\text{shift and scale})$$

and for the difference of the two arms' T 's, the variances add while the biases *subtract before squaring*:

$$\text{Var}^\theta(\tilde{\delta}) = \frac{n_T\theta_T(1-\theta_T)}{(n_T+2)^2} + \frac{n_P\theta_P(1-\theta_P)}{(n_P+2)^2} \quad (\text{independent arms})$$

$$b_{\tilde{\delta}}(\theta) = \frac{1-2\theta_T}{n_T+2} - \frac{1-2\theta_P}{n_P+2} \quad (\text{bias of a difference})$$

$$R_{\tilde{\delta}}(\theta) = \boxed{\frac{n_T\theta_T(1-\theta_T)}{(n_T+2)^2} + \frac{n_P\theta_P(1-\theta_P)}{(n_P+2)^2} + \left(\frac{1-2\theta_T}{n_T+2} - \frac{1-2\theta_P}{n_P+2}\right)^2} \quad (\text{bias-variance})$$

The subtraction is worth a sentence, because it is the one place a plausible shortcut fails: the risk of the difference is *not* the sum of the two arms' mean squared errors—expanding the square produces the cross term $-2b_T(\theta_T)b_P(\theta_P)$, which vanishes only

when one of the biases does. With equal arm sizes the two biases cancel entirely along the diagonal $\theta_T = \theta_P$, where the shading is a pure variance gain; along the anti-diagonal $\theta_P = 1 - \theta_T$ they reinforce, and the rate difference wins near the corners. Figure 1 shows both slices; the risk surfaces cross, so neither rule dominates the other (notes, Supplements 1.A.3–1.A.4).

The Bayes risks are the prior expectations of the risk functions (notes, Def. 1.6.2; HPS, Defs. 3–4, p. 9). Under the uniform prior, three moments do all the work:

$$\int_0^1 \theta(1-\theta) d\theta = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}, \quad \int_0^1 (1-2\theta) d\theta = 0, \quad \int_0^1 (1-2\theta)^2 d\theta = 1 - 2 + \frac{4}{3} = \frac{1}{3}.$$

For the rate difference, integrating the risk term by term,

$$r_{\hat{\delta}} = E\{R_{\hat{\delta}}(\theta)\} = \boxed{\frac{1}{6} \left(\frac{1}{n_T} + \frac{1}{n_P} \right)} \quad (\text{integrate the risk; first moment above})$$

For the shaded difference, the squared bias expands as $b_T^2 + b_P^2 - 2b_T b_P$, and the cross term integrates to zero because the priors are independent and $\int_0^1 (1-2\theta) d\theta = 0$; each arm then contributes

$$\frac{n \cdot \frac{1}{6} + \frac{1}{3}}{(n+2)^2} = \frac{(n+2)/6}{(n+2)^2} = \frac{1}{6(n+2)},$$

so that

$$r_{\tilde{\delta}} = \boxed{\frac{1}{6} \left(\frac{1}{n_T + 2} + \frac{1}{n_P + 2} \right)} \\ (\text{independent priors; second and third moments above})$$

strictly smaller than $r_{\hat{\delta}}$, for every pair of arm sizes.

The margin is no accident, and the class example is built to expose why. Under the uniform prior the posterior density of one arm's θ given $X = x$ is proportional to $\theta^x(1 - \theta)^{n-x} = \theta^{(1+x)-1}(1 - \theta)^{(1+n-x)-1}$ —the $\text{Beta}(1 + x, 1 + n - x)$ kernel—whose mean, by the Beta facts of the lecture (notes, Remark 1.8.5), is exactly $(1 + x)/(n + 2)$: the shaded estimator *is* the posterior mean, and the posterior mean is the Bayes rule for squared-error loss (notes, Supplement 1.A.5; HPS, Thm. 2, p. 38; DeGroot and Schervish, Cor. 7.4.1, p. 410). Since the arms' posteriors are independent, the difference of the posterior means is the Bayes rule for the difference, so $r_{\hat{\delta}}$ is not merely smaller than $r_{\hat{\gamma}}$ —it is the smallest Bayes risk any decision rule can achieve here, and it equals the expected posterior variance, $1/(6(n_T + 2)) + 1/(6(n_P + 2))$, arm by arm. The computation is classical on every shelf: HPS §2.8 carries it out for a general $\text{Beta}(\alpha, \beta)$ prior, displaying the posterior mean $(\alpha + S)/(\alpha + \beta + n)$ and the uniform special case $(S + 1)/(n + 2)$ (Eq. (27) and following, p. 39); DeGroot and Schervish state the updating rule as Theorem 7.3.1 (pp. 394–395) and the estimator as Example 7.4.3 (p. 410); Blitzstein and Hwang tell it as the Beta-Binomial conjugacy story—“the Beta is the conjugate prior of the Binomial” (Story 8.3.3, pp. 383–385); and Pitman derives the same posterior from a uniform prior and lands on $E(\Pi \mid S_n = k) = (k + 1)/(n + 2)$ (pp. 420–421)—Laplace's rule of succession. The shaded estimator has been in service for two centuries.

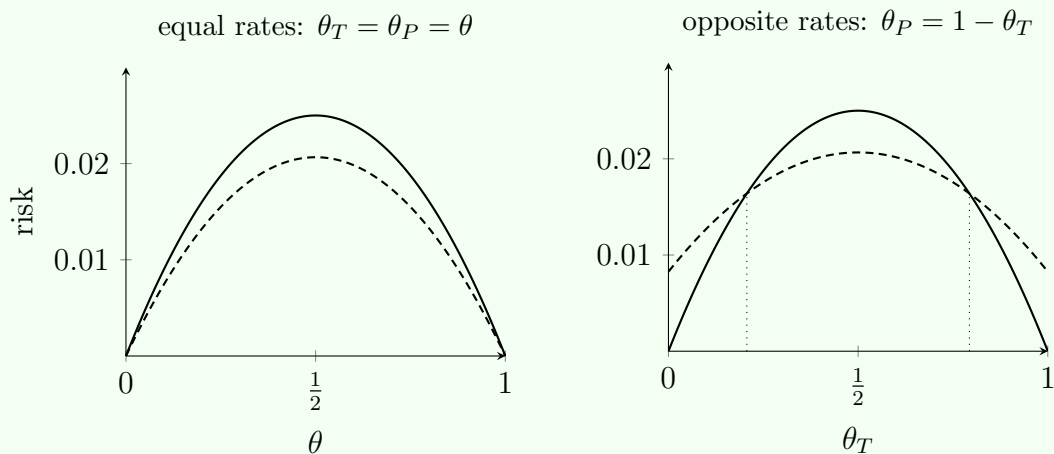


Figure 1. TWO SLICES OF THE RISK SURFACES. The risks of $\hat{\delta}$ (solid) and $\tilde{\delta}$ (dashed) live on the square $(\theta_T, \theta_P) \in [0, 1]^2$; both panels use $n_T = n_P = 20$. Along the diagonal (left) the two shading biases cancel and the shaded rule is a pure variance gain, uniformly better. Along the anti-diagonal (right) the biases reinforce, and the rate difference wins outside $\theta_T \in (0.21, 0.79)$ (dotted). The surfaces cross: neither rule dominates.

■

Problem 2

Consider a new problem, this time we have a sample (sample means independent and identically distributed) of n normal variables with unknown expectation φ and known variance σ^2 . Assume that you want to estimate the expectation and that you are using squared error loss. Suppose your prior is that the expectation is normal with expectation 1 and variance 1. Let S denote the sum of the observed values, and consider two estimates: S/n and $(1-a)S/n + a$ for some fixed a . What are the risk functions and Bayes risk for these estimators (for the second estimator, your answer will depend on a)? Also, what value of a minimizes the Bayes Risk?

SOLUTION:

Let $Y_1, \dots, Y_n$ be independent and identically distributed $N(\varphi, \sigma^2)$ with σ^2 known, $S = \sum_i Y_i$, and write $d_0 = S/n$ (the sample mean) and $d_a = (1-a)S/n + a$. The anatomy of d_a is the same as Problem 1's shaded rule: it pulls the sample mean toward 1—which is exactly the prior mean—with weight a , and the family sweeps from no shrinkage ($a = 0$, recovering d_0) to ignoring the data entirely ($a = 1$, reporting the prior mean).

The risks are two short bias–variance computations. Since $E^\varphi\{S\} = n\varphi$ and $\text{Var}^\varphi(S) = n\sigma^2$ (independent, identically distributed summands),

$$\begin{aligned} E^\varphi\{d_0\} &= \frac{n\varphi}{n} = \varphi && \text{(linearity)} \\ \text{Var}^\varphi(d_0) &= \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} && \text{(scaling)} \\ R_{d_0}(\varphi) &= \boxed{\frac{\sigma^2}{n}} && \text{(unbiased; bias–variance decomposition)} \end{aligned}$$

a *constant* risk: the flat line of Figure 2, the same at every candidate truth. For the shrunk rule,

$$\mathbb{E}^\varphi\{d_a\} = (1 - a)\varphi + a, \quad b_a(\varphi) = a(1 - \varphi) \quad (\text{linearity; algebra})$$

$$\text{Var}^\varphi(d_a) = (1 - a)^2 \frac{\sigma^2}{n} \quad (\text{shift and scale})$$

$$R_{d_a}(\varphi) = \boxed{(1 - a)^2 \frac{\sigma^2}{n} + a^2(1 - \varphi)^2} \quad (\text{bias-variance decomposition})$$

a parabola in φ with vertex at the prior mean: d_a buys a variance reduction by the factor $(1 - a)^2$ at the price of a bias proportional to the distance of the truth from 1. For $0 < a \leq 1$ the parabola dips below the flat line exactly on the window $(1 - \varphi)^2 < (2 - a)\sigma^2/(an)$ and rises above it outside, so—as in Problem 1, and as always in this course so far—the risk curves cross and neither rule dominates.

The Bayes risks are the prior expectations of these risk functions, under $\varphi \sim N(1, 1)$. The only prior moment needed is $\mathbb{E}\{(1 - \varphi)^2\} = \text{Var}(\varphi) + (1 - \mathbb{E}\{\varphi\})^2 = 1 + 0 = 1$, and so

$$r_{d_0} = \boxed{\frac{\sigma^2}{n}} \quad (\text{expectation of a constant})$$

$$r(a) = \mathbb{E}\{R_{d_a}(\varphi)\} = \boxed{(1 - a)^2 \frac{\sigma^2}{n} + a^2} \quad (\text{integrate the risk; prior mean and variance})$$

To minimize over a , complete the square. Writing $v = \sigma^2/n$,

$$r(a) = (1 + v)a^2 - 2va + v \quad (\text{expand})$$

$$= (1 + v) \left(a - \frac{v}{1 + v} \right)^2 + \frac{v}{1 + v} \quad (\text{complete the square})$$

so the Bayes risk is minimized at

$$\boxed{a^* = \frac{v}{1+v} = \frac{\sigma^2}{n+\sigma^2}} \quad \text{with} \quad r(a^*) = \frac{\sigma^2}{n+\sigma^2},$$

and every a strictly between 0 and $2a^*$ already improves on the sample mean, the parabola $r(a)$ being symmetric about a^* (Figure 2). That a^* and $r(a^*)$ share one closed form is an accident of the unit prior variance: for a general prior variance v_0^2 the weight is $a^* = (\sigma^2/n)/(v_0^2 + \sigma^2/n)$ while the minimized Bayes risk is $r(a^*) = v_0^2(\sigma^2/n)/(v_0^2 + \sigma^2/n)$, and the two separate.

As in Problem 1, the optimal member of the family is no stranger. With weight a^* ,

$$d_{a^*} = \frac{n}{n+\sigma^2} \cdot \frac{S}{n} + \frac{\sigma^2}{n+\sigma^2} \cdot 1,$$

the precision-weighted average of the data and the prior: the prior carries precision 1 (variance one), the sample mean carries precision n/σ^2 , and a^* is precisely the prior's share of the total precision. This is the posterior mean of the normal-normal model, so d_{a^*} is the Bayes rule and $r(a^*) = \sigma^2/(n+\sigma^2)$ is the posterior variance—the smallest Bayes risk available to any rule, not just to this family. The course text works this exact problem: HPS §2.8 posits a normal prior $N(\mu_0, \alpha^2)$, displays the posterior mean $(\sigma^2\mu_0 + \alpha^2n\bar{x})/(\sigma^2 + \alpha^2n)$ and its “more useful version” as a weighted average of μ_0 and $\bar{x}$, computes the posterior variance $\alpha^2\sigma^2/(\sigma^2 + \alpha^2n)$, and even reports the Bayes-risk ratio of the posterior mean to the sample mean, $1/(1+\delta)$ with $\delta = \sigma^2/(n\alpha^2)$ (pp. 39–41)—all of which specialize, at $\mu_0 = \alpha^2 = 1$, to the boxed formulas above. DeGroot and Schervish state the conjugate updating as Theorem 7.3.3 with the same weighted-average reading (Eq. (7.3.3), pp. 398–400); Blitzstein and Hwang put it in precision language—“the posterior precision of θ is the sum of the prior precision and

the data precision” (Example 12.1.8, pp. 544–546); Ross reaches the same posterior through the bivariate normal (Example 8b, p. 347).

Taken together, the two problems are one lesson told in two conjugate families. In each, a family of shrinkage rules connects the unbiased estimator to the prior mean; the risk curves always cross, so risk alone cannot choose; the prior’s verdict picks the member that shrinks by exactly the prior’s share of the total precision; and that member turns out to be the posterior mean, with the minimized Bayes risk equal to the expected posterior variance— $\frac{1}{6}(\frac{1}{n_T+2} + \frac{1}{n_P+2})$ in the Beta–binomial trial, $\sigma^2/(n + \sigma^2)$ here.

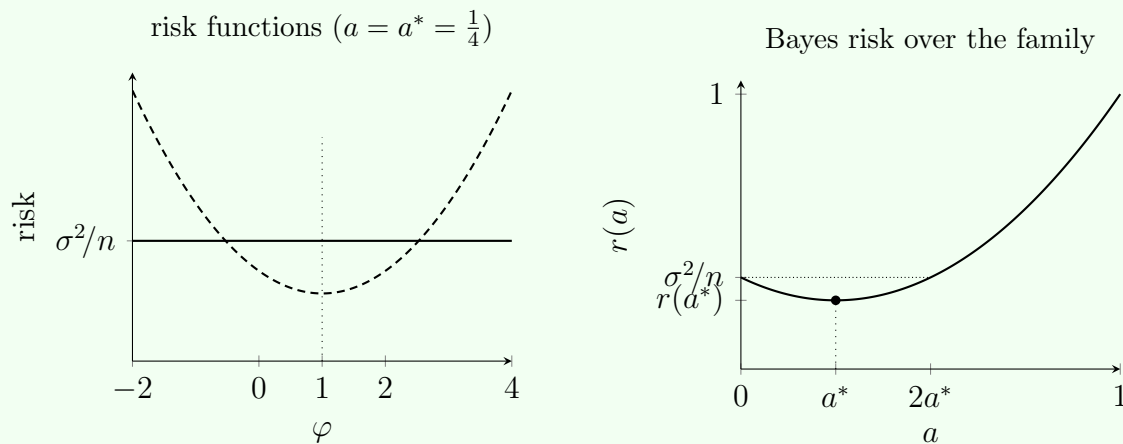


Figure 2. THE SHRINKAGE FAMILY AND ITS BEST MEMBER ($\sigma^2 = 4$, $n = 12$, so $\sigma^2/n = \frac{1}{3}$ and $a^* = \frac{1}{4}$). Left: the sample mean’s risk is flat; the shrunk rule’s risk is a parabola with vertex at the prior mean $\varphi = 1$ (dotted)—better nearby, worse far away, neither dominating. Right: the Bayes risk over the family, minimized at $a^* = \sigma^2/(n + \sigma^2)$ where d_a becomes the posterior mean; every $a \in (0, 2a^*)$ improves on the sample mean (dotted level).



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